Informatica

Informatica® PowerExchange for Netezza
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User Guide for PowerCenter

Informatica PowerExchange for Netezza User Guide for PowerCenter
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Preface

Use the *Informatica® PowerExchange® for Netezza User Guide* to learn how to read from or write to Netezza by using the PowerCenter Client. Learn to create a Netezza connection, develop mappings, parameterize Netezza sources and targets, and run sessions in an Informatica domain.

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CHAPTER 1

Introduction to PowerExchange for Netezza

This chapter includes the following topics:

- PowerExchange for Netezza Overview, 7
- Code Pages, 7

PowerExchange for Netezza Overview

PowerExchange for Netezza provides bidirectional connectivity between PowerCenter and IBM Netezza Platform Software to read and write data.

The Designer uses a relational connector to connect to the Netezza database. You can import Netezza tables as sources and target definitions. You can connect to the Netezza Performance Server to read data from Netezza tables and load data to Netezza tables. The Netezza Performance Server integrates database, server, and storage in a single system.

Configure a Netezza database connection in a mapping to read data from and write to Netezza.

Code Pages

When the PowerCenter Integration Service runs in Unicode mode, it encodes Netezza data of the Nchar(m) and NVarchar(m) datatypes in UTF-8. It encodes Netezza data of the Varchar and Char datatypes in Latin-9.

If the data contains extended ASCII characters or UTF-8 characters, run the PowerCenter Integration Service in Unicode mode.

CHAPTER 2

PowerExchange for Netezza Configuration

This chapter includes the following topics:

- PowerExchange for Netezza Configuration Overview, 8
- Prerequisites, 8
- Registering the Plug-in, 9
- Connecting to a Netezza Database from Windows, 9
- Connecting to a Netezza Database, 10
- Upgrading PowerExchange for Netezza, 12

PowerExchange for Netezza Configuration Overview

PowerExchange for Netezza installs with PowerCenter.

To configure PowerExchange for Netezza, complete the following steps:

1. Complete the prerequisites.
2. Register the PowerExchange for Netezza plug-in if you want to read or write data in bulk mode. To read or write data in normal mode, you do not need to perform configuration steps.

Prerequisites

Before you configure PowerExchange for Netezza, complete the following tasks:

- Install or upgrade PowerCenter.
- Install the client and server components of the Netezza Performance Server.
- Verify that the Netezza database user has the following privileges on the database:
  - CREATE TABLE
  - CREATE EXTERNAL TABLE
  - DELETE

- DROP
- INSERT
- LIST
- SELECT
- TRUNCATE
- UPDATE

Registering the Plug-in

To read or write Netezza data in bulk mode, you need to register the plug-in with the repository. You do not need to register the plug-in with the repository to read or write Netezza data in normal mode.

To register the plug-in, the repository must be running in exclusive mode. Use the Informatica Administrator or the *pmrep* RegisterPlugin command to register the plug-in.

The plug-in file for PowerExchange for Netezza is `pmnetezza.xml`. When you install the Service component, the installer copies `pmnetezza.xml` to the following directory:

```
<PowerCenter Installation Directory>\server\bin\Plugin
```

Note: If you do not have the correct privileges to register the plug-in, contact the user who manages the PowerCenter Repository Service.

Connecting to a Netezza Database from Windows

Install and configure ODBC on the machines where the PowerCenter Integration Service process runs and where you install the PowerCenter Client. You must configure connectivity to the following Informatica components on Windows:

- **PowerCenter Integration Service.** Install the Netezza ODBC driver on the machine where the PowerCenter Integration Service process runs. Use the Microsoft ODBC Data Source Administrator to configure ODBC connectivity.
- **PowerCenter Client.** Install the Netezza ODBC driver on each PowerCenter Client machine that accesses the Netezza database. Use the Microsoft ODBC Data Source Administrator to configure ODBC connectivity. Use the Workflow Manager to create a database connection object for the Netezza database.

Configuring ODBC Connectivity

You can configure ODBC connectivity to a Netezza database.

The following steps provide a guideline for configuring ODBC connectivity. For specific instructions, see the database documentation.

1. Create an ODBC data source for each Netezza database that you want to access.

   To create the ODBC data source, use the driver provided by Netezza.

   Create a System DSN if you start the Informatica service with a Local System account logon. Create a User DSN if you select the This account log in option to start the Informatica service.

   After you create the data source, configure the properties of the data source.
2. Enter a name for the new ODBC data source.
3. Enter the IP address/host name and port number for the Netezza server.
4. Enter the name of the Netezza schema where you plan to create database objects.
5. Configure the path and file name for the ODBC log file.
6. Verify that you can connect to the Netezza database.

   You can use the Microsoft ODBC Data Source Administrator to test the connection to the database. To test the connection, select the Netezza data source and click Configure. On the Testing tab, click Test Connection and enter the connection information for the Netezza schema.

Connecting to a Netezza Database

Install and configure Netezza ODBC driver on the machine where the PowerCenter Integration Service process runs. Use the DataDirect Driver Manager in the DataDirect driver package shipped with the Informatica product to configure the Netezza data source details in the odbc.ini file.

Configuring ODBC Connectivity

You can configure ODBC connectivity to a Netezza database.

The following steps provide a guideline for configuring ODBC connectivity. For specific instructions, see the database documentation.

1. To configure connectivity for the integration service process, log in to the machine as a user who can start a service process.
2. Set the ODBCHOME, NZ_ODBC_INI_PATH, and PATH environment variables.

   ODBCHOME. Set the variable to the ODBC installation directory. For example:

   Using a Bourne shell:

   ```
   $ ODBCHOME=<Informatica server home>/ODBC7.1; export ODBCHOME
   ```

   Using a C shell:

   ```
   $ setenv ODBCHOME =<Informatica server home>/ODBC7.1
   ```

   PATH. Set the variable to the ODBCHOME/bin directory. For example:

   Using a Bourne shell:

   ```
   PATH="${PATH}:$ODBCHOME/bin"
   ```

Using a C shell:

```
$ setenv PATH ${PATH}:$ODBCHOME/bin
```

NZ_ODBC_INI_PATH. Set the variable to point to the directory that contains the odbc.ini file. For example, if the odbc.ini file is in the $ODBCHOME directory:

Using a Bourne shell:

```
NZ_ODBC_INI_PATH=$ODBCHOME; export NZ_ODBC_INI_PATH
```

Using a C shell:

```
$ setenv NZ_ODBC_INI_PATH $ODBCHOME
```

3. Set the shared library environment variable.

   The shared library path must contain the ODBC libraries. It must also include the Informatica services installation directory `(server_dir)`.

   Set the shared library environment variable based on the operating system. Set the Netezza library folder to <NetezzaInstallationDir>/lib64.

   The following table describes the shared library variables for each operating system:

Operating System	Variable
Linux	LD_LIBRARY_PATH
AIX	LIBPATH

   For example, use the following syntax for Linux:

   - Using a Bourne shell:

     ```
     $ LD_LIBRARY_PATH="${LD_LIBRARY_PATH}:$HOME/server_dir:$ODBCHOME/
     lib:<NetezzaInstallationDir>/lib64”
     export LD_LIBRARY_PATH
     ```

   - Using a C shell:

     ```
     $ setenv LD_LIBRARY_PATH "${LD_LIBRARY_PATH}:$HOME/server_dir:$ODBCHOME/
     lib:<NetezzaInstallationDir>/lib64"
     ```

   For AIX

   - Using a Bourne shell:

     ```
     $ LIBPATH=${LIBPATH}:$HOME/server_dir:$ODBCHOME/lib:<NetezzaInstallationDir>/
     lib64; export LIBPATH
     ```

   - Using a C shell:

     ```
     $ setenv LIBPATH ${LIBPATH}:$HOME/server_dir:$ODBCHOME/
     lib:<NetezzaInstallationDir>/lib64
     ```

4. Edit the existing odbc.ini file or copy the odbc.ini file to the home directory and edit it.

   This file exists in $ODBCHOME directory.

   ```
   $ cp $ODBCHOME/odbc.ini $HOME/.odbc.ini
   ```

   Add an entry for the Netezza data source under the section [ODBC Data Sources] and configure the data source.

   For example:

   ```
   [NZSQL]
   Driver = /export/home/appsqa/thirdparty/netezza/lib64/libnzodbc.so
   Description = NetezzaSQL ODBC
   Servername = netezza1.informatica.com
   Port = 5480
   Database = infa
   ```

```
Username = admin
Password = password
Debuglogging = true
StripCRLF = false
PreFetch = 256
Protocol = 7.0
ReadOnly = false
ShowSystemTables = false
Socket = 16384
DateFormat = 1
TranslationDLL =
TranslationName =
TranslationOption =
NumericAsChar = false
```

For more information about Netezza connectivity, see the Netezza ODBC driver documentation.

5. Verify that the last entry in the odbc.ini file is InstallDir and set it to the ODBC installation directory.

   For example:

   ```
   InstallDir=<Informatica install directory>/<ODBCHOME directory>
   ```

6. Edit the .cshrc or .profile file to include the complete set of shell commands.
7. Restart the Informatica services.

Upgrading PowerExchange for Netezza

To upgrade PowerExchange for Netezza from earlier versions, complete the prerequisite tasks. Update the PowerExchange for Netezza plug-in registration if you want to read or write data in bulk mode.

CHAPTER 3

Netezza Sources and Targets

This chapter includes the following topics:

- Netezza Sources and Targets Overview, 13
- Source Qualifier Properties, 13
- Importing Netezza Source Definitions, 14
- Importing Netezza Target Definitions, 14

Netezza Sources and Targets Overview

Netezza source and target definitions represent metadata for Netezza tables. When you import Netezza definitions, you can choose to preview data in the tables.

You can edit definitions to configure the properties that you did not import from Netezza. If you want to enforce key constraints, define them in the Designer. When you run a session, the PowerCenter Integration Service establishes relationships within the pipeline based on source and target definitions. Netezza does not enforce key constraints.

Source Qualifier Properties

You can configure source qualifier properties to sort the number of input ports and to retrieve distinct data from a Netezza source. You can override the values in the session properties.

The following table describes the source qualifier properties:

Source Options	Description
Select Distinct	Selects unique values. Netezza ignores trailing spaces. Therefore, the PowerCenter Integration Service might extract fewer rows than expected.
Source Filter	Reduces the number of rows the PowerCenter Integration Service queries. Use the following syntax: `<table name>."<field name>" <operator> <value>` The filter condition is case sensitive.

Source Options	Description
Number of Sorted Ports	Number of columns used when sorting rows queried from the source. The PowerCenter Integration Service adds an ORDER BY clause to the default query when it reads source rows. The ORDER BY clause includes the number of ports specified, starting from the top of the transformation. When you specify the number of sorted ports, the database sort order must match the session sort order. Default is 0.
SQL Query	Overrides the default query. Enclose column names in double quotes. The SQL query is case sensitive.

Importing Netezza Source Definitions

To create a Netezza source definition, use the Source Analyzer to import source metadata with the Netezza relational data source.

1. In the Source Analyzer, click **Sources** > **Import from Database.**
2. Select the Netezza data source used to connect to the source database.

   If you need to create or modify a Netezza data source, click the **Browse** button to open the ODBC Administrator. Create the Netezza data source and click **OK.** Select the new Netezza data source.
3. Enter a database user name and password to connect to the database.

   Note: The user must have the appropriate database permissions to view the object.

   You may need to specify the owner name for database objects you want to use as sources.
4. Optionally, use the search field to limit the number of tables that appear.
5. Click **Connect.**

   If no table names appear, or if the table you want to import does not appear, click **All**.
6. Scroll down through the list of sources to find the source you want to import. Select the relational object or objects you want to import.

   You can hold down the Shift key to select a block of sources within one folder or hold down the Ctrl key to make non-consecutive selections within a folder. You can also select all tables within a folder by selecting the folder and clicking **Select All.** Use the **Select None** button to clear all highlighted selections.
7. Click **OK.**

The source definition appears in the Source Analyzer. In the Navigator, the source definition appears in the Sources node of the active repository folder under the source database name.

Importing Netezza Target Definitions

To create a Netezza target definition, use the Target Designer to import source metadata with the Netezza relational data source.

1. In the Target Designer, click **Targets** > **Import from Database.**

2. Select the Netezza data source used to connect to the target database.

   If you need to create or modify a Netezza data source, click the **Browse** button to open the ODBC Administrator. Create the Netezza data source and click **OK**. Select the new Netezza data source.

3. Enter the user name and password to connect to the database, and click **Connect**.

   If you are not the owner of the table you want to use as a target, specify the owner name.

4. Drill down through the list of database objects to view the available tables as targets.

5. Select the relational table or tables to import the definitions into the repository.

   You can hold down the Shift key to select a block of tables, or hold down the Ctrl key to make non-consecutive selections. You can also use the **Select All** and **Select None** buttons to select or clear all available targets.

6. Click **OK.**

   The selected target definitions appear in the Navigator under the Targets node.

CHAPTER 4

Netezza Sessions and Workflows

This chapter includes the following topics:

- Data Transfer Modes in Netezza, 16
- PowerExchange for Netezza Connections, 18
- Session Configuration with a Netezza Source, 19
- Parameterization for Netezza Sources, 19
- Session Configuration with a Netezza Target, 20
- Parameterization for Netezza Targets, 24
- Netezza Target Data Update, 24
- Netezza Session Configuration for Optimal Performance, 26
- Troubleshooting Netezza Sessions, 27

Data Transfer Modes in Netezza

You can transfer data in Netezza by using normal and bulk mode.

Normal Mode

In normal mode, the PowerCenter Integration Service extracts and loads data row by row.

Bulk Mode

You can transfer data in Netezza by using bulk mode. Use bulk mode to increase session performance.

In bulk mode, the PowerCenter Integration Service reads and writes Netezza data through an external table. An external table definition is stored within the Netezza database but the data is saved externally in a location that is accessible to the Netezza host or the client system. Create external tables to structure your loading operation and manipulate data by using Netezza SQL.

When the PowerCenter Integration Service extracts from Netezza, it creates an external table in the pipe directory path specified for extraction. When the PowerCenter Integration Service loads to Netezza, it creates an external table in the pipe directory path specified for loading. You can load data from the external table to the target. If duplicate row handling is configured, data is loaded from the external table to a temporary table and then finally to the target.

To transfer data in Netezza in bulk mode, complete the following steps:

1. Verify that the Netezza database user has the LIST and CREATE EXTERNAL TABLE privileges on the database.
2. Register the PowerExchange for Netezza plug-in with the repository.
3. Configure the session to use a Netezza bulk reader and Netezza bulk writer.
4. Configure the session properties as described in the following sections.

Normal and Bulk Mode Features

The following table lists the features supported in normal and bulk modes:

Feature	Normal Mode	Bulk Mode
Source and target table name override	Supported	Supported
Source and target schema name override	Supported	Supported
Recovery	Supported	Unsupported
Real-time sessions	Supported	Unsupported
Session on grid	Supported	Limited support
Commit interval	Supported	Unsupported
Implicit join based on primary key and foreign key	Supported	Unsupported
Pushdown optimization	Supported	Unsupported
Duplicate row handling	Unsupported	Supported
Pre-SQL	Supported	Supported
Post-SQL	Supported	Supported
Truncate target table	Supported. Run the delete command. If you specify an SQL statement in the Pre-SQL property, the PowerCenter Integration Service runs the SQL statement after the data in the table is deleted.	Supported. Run the truncate table command. If you specify an SQL statement in the Pre-SQL property, the PowerCenter Integration Service runs the SQL statement before the table is truncated.
Target connection group	Supported	Limited support. Follow the criteria, and rules and guidelines applicable for using target connection groups in bulk mode.
Partitioning	Database, hash, key range, pass-through, and round-robin partitioning are supported.	Pass-through partitioning is supported.

PowerExchange for Netezza Connections

Use a relational connection object for each Netezza source or target that you want to access.

The relational database connection defines how the PowerCenter Integration Service accesses the underlying database for Netezza Performance Server. When you configure a Netezza connection, you specify the connection attributes that the PowerCenter Integration Service uses to connect to Netezza.

Netezza Connection Properties

The following table describes the Netezza connection properties that you must configure:

Property	Description
User Name	Database user name with the appropriate read and write database permissions to access Netezza Performance Server.
Use Parameter in Password	Indicates the password for the database user name is a session parameter, $Param*Name*. Define the password in the workflow or session parameter file, and encrypt it by using the *pmpasswd* CRYPT_DATA option. Default is disabled.
Password	Password for the database user name.
Connect String	ODBC data source to connect to Netezza Performance Server.
Code Page	Code page associated with Netezza Performance Server.
Connection Environment SQL	Runs an SQL command with each database connection. Default is disabled.
Transaction Environment SQL	Runs an SQL command before the initiation of each transaction. Default is disabled.
Connection Retry Period	Number of seconds the PowerCenter Integration Service attempts to reconnect to the database if the connection fails. If the PowerCenter Integration Service cannot connect to the database in the retry period, the session fails. Default value is 0.

Creating a Netezza Connection

Create a Netezza connection before you run a Netezza session.

1. In the Workflow Manager, connect to a repository.
2. Click **Connections** > **Relational**.

   The **Relational Connection Browser** dialog box appears.
3. Click **New**.

   The **Select Subtype** dialog box appears.
4. Select **Netezza** from the **Select Subtype** list.
5. Click **OK**.

   The **Connection Object Definition** dialog box appears.
6. Enter the connection properties.

7. Click **OK**.

   The Netezza connection appears in the Connection Browser list.

Session Configuration with a Netezza Source

You can configure the session properties for a Netezza source on the Mapping tab. Define the properties for each source instance in the session.

To extract data in normal mode, configure the session to use a relational reader. To extract data in bulk mode, configure the session to use a Netezza bulk reader. The session properties for normal mode are the same as that of any other relational source.

The following table describes the session properties that you must configure to extract Netezza source data in bulk mode:

Attribute Name	Description
Socket Buffer Size	Set the socket buffer size to 25 to 50 % of the DTM buffer size to increase session performance. You might need to test different settings for optimal performance. Enter a value between 4096 and 2147483648 bytes. Default is 8388608 bytes.
Pipe Directory Path	Path for the PowerCenter Integration Service to create the pipe for the external table. If you do not specify the path, the PowerCenter Integration Service uses the following directory to create the pipe for the external table: `<PowerCenter Installation Directory>/server/bin`
Delimiter	Delimiter separates successive input fields. You can enter any value supported by the Netezza Performance Server. The value can be a part of the data for the Netezza source. Default is |.
NullValue	NullValue parameter of an external table. The PowerCenter Integration Service uses the NullValue internally. Maximum value is one character. Default is blank.
EscapeCharacter	Escape character of an external table. If the data contains NULL, CR, and LF characters in the Char or Varchar field, you need to add an escape character in the source data before extracting. Enter an escape character before the data. The supported escape character is backslash (\).

Note: You can view load statistics in the session log. The load summary in the Workflow Monitor does not display load statistics.

Parameterization for Netezza Sources

You can parameterize Netezza session properties to override the session properties during run time.

For example, you want to extract data from a customers table. The customers table has multiple schemas that contain customer information for different countries. You want to use one mapping to extract data from multiple schemas based on the country name instead of creating different mappings for different schemas. In this case, you can parameterize the owner name and source table name in the session properties, and use the same mapping to extract data from multiple schemas.

In normal mode, you can parameterize the following session properties for Netezza sources:

- Owner Name
- Pre SQL
- Post SQL
- Source Filter
- Source Table Name
- SQL Query
- User Defined Join

In bulk mode, you can parameterize the following session properties for Netezza sources:

- Delimiter
- EscapeCharacter
- NullValue
- Owner Name
- Pre SQL
- Post SQL
- Source Filter
- Source Table Name
- SQL Query
- User Defined Join

Session Configuration with a Netezza Target

You can configure target properties for a session that writes data to Netezza targets:

- Target database connection
- Target properties
- Update strategy
- Multiple targets referring to the same table
- Pipeline partitioning

Target Properties

You can configure the session properties for Netezza targets in the Transformations view on the Mapping tab. Define the properties for each target instance in the session.

To load data in normal mode, configure the session to use a relational writer. To load data in bulk mode, configure the session to use a Netezza bulk writer. The session properties for normal mode are the same as that of any other relational target.

The following table describes the session properties that you must configure to load Netezza target data in bulk mode:

Target Property	Description
Socket Buffer Size	Set the socket buffer size to 25 to 50 % of the DTM buffer size to increase session performance. You might need to test different settings for optimal performance. Enter a value between 4096 and 2147483648 bytes. Default is 8388608 bytes.
Pipe Directory Path	Path for the PowerCenter Integration Service to create the pipe for the external table. If you do not specify the path, the PowerCenter Integration Service uses the following directory to create the pipe for the external table: `<PowerCenter Installation Directory>/server/bin`

Error Log Directory Name	Error log directory can reside on the machine where the PowerCenter Integration Service runs. For example, you can use the following directory:
	`$PMBadFileDir`
	By default, the PowerCenter Integration Service creates the error log in the following directory on the machine hosting the Netezza Performance Server:
	`/tmp`
	The PowerCenter Integration Service creates a bad file in the error log directory if the data is not valid.

Truncate Target Table Option	The PowerCenter Integration Service truncates the target before loading. Run the truncate table command. Default is disabled. If you specify an SQL statement in the Pre-SQL property, the PowerCenter Integration Service runs the SQL statement before the table is truncated.
Target Table Name	You can override the default target table name.
Delimiter	Set the delimiter to any value supported by the Netezza Performance Server. The delimiter separates successive input fields. The value must not be a part of the input data. Default is |.
Control Character	CTRLCHARS parameter of the external table to transfer data containing control characters. You can enter control characters for Char and Varchar fields. If you enter a control character, you must add an escape character for the NULL, CR, and LF fields. Default is TRUE.
CRINSTRING	CRINSTRING parameter to transfer data containing carriage returns (CR). You can enter a non escape CR in Char or Varchar fields. To load the control characters present in the Char and Varchar fields, set the CTRLCHARS and CRINSTRING parameters to TRUE in the session properties for the Netezza source. Default is TRUE.
NullValue	NullValue parameter of the external table. The PowerCenter Integration Service uses the NullValue internally. Maximum value is one character. Default is blank.
EscapeCharacter	Escape character of the external table. If the data contains NULL, CR, and LF characters in the Char or Varchar field, you need to add an escape character for these fields before loading. Enter a backslash (\) as the escape character.
Quoted Value	QUOTEDVALUE parameter of the external table. Select SINGLE or DOUBLE to enclose the field in single or double quotes. Select NO to omit quotes. Default is NO. The quoted value is not a part of the data.

Target Property	Description
Ignore Key Constraints	Ignores constraints on primary key fields. When you select this option, the PowerCenter Integration Service can write duplicate rows with the same primary key to the target. Default is disabled. The PowerCenter Integration Service ignores this value when the target operation is “update as update” or “update else insert.”
Duplicate Row Handling Mechanism	Determines how the PowerCenter Integration Service handles duplicate rows. Select one of the following values: - First Row. The PowerCenter Integration Service passes the first row to the target and rejects the rows that follow with the same primary key. - Last Row. The PowerCenter Integration Service passes the last duplicate row to the target and discards the rest of the rows. Default is First Row.

Null Values and Empty Strings

The target may contain null values even if you configure a column in the source definition to be not null. The PowerCenter Integration Service loads empty strings as null values to the target.

Unprojected Columns

When the PowerCenter Integration Service generates SQL to load to a Netezza target, it ignores target columns that are not connected in the mapping. If a default value is defined in Netezza for an unconnected column, Netezza updates or populates the column with the default value.

Pipeline Partitioning

You can increase the number of partitions in a pipeline to improve session performance. When you increase the number of partitions, the PowerCenter Integration Service can create multiple connections to sources and targets and process partitions of sources and target data concurrently.

The Netezza Performance Server divides data into data slices. In a partitioned session that reads data from Netezza, each partition reads a different data slice to prevent data duplication except in the following cases:

- You enter an SQL override query for a partition.
- You enter different values for the source filter across partitions.
- You enter different values for the user-defined join across partitions.

Rules and Guidelines for Pipeline Partitioning

Use the following rules and guidelines when you configure multiple partitions in a Netezza session:

- If you load in bulk mode, use pass-through partitioning. If you load in normal mode, you can use database, hash, key range, pass-through, or round-robin partitioning.
- Verify that the session properties for delete and update on the Mapping tab are not enabled for more than one partition. You cannot perform multiple updates, multiple deletes, or update and delete simultaneously on a Netezza target.
- To avoid unpredictable session results, configure the session properties for insert, delete, update, and duplicate row handling to have the same value for each partition.

- If you run a partitioned session that joins multiple sources, link the first column in the Source Qualifier transformation to a source column that represents data for the Netezza table with the best distribution in Netezza. This means that the Netezza table is more uniformly distributed across Snippet Processing Units (SPU) than other tables.
- If you run a partitioned session with key constraints, only one partition shows load statistics.

Target Connection Groups

A target connection group is a group of targets that the PowerCenter Integration Service uses to determine commits and loading. When the PowerCenter Integration Service writes to Netezza, it commits data in the same transaction for all targets in a target connection group. When the PowerCenter Integration Service needs to perform a rollback, the PowerCenter Integration Service rolls back all targets in the target connection group.

Criteria for using Target Connection Groups in Bulk Mode

When you load data in bulk mode, all Netezza targets in the same target connection group must meet the following criteria:

- Belong to the same pipeline.
- Belong to the same partition.
- Have the same database connection name, user name, and password.

Rules and Guidelines for using Target Connection Groups in Bulk Mode

The following table describes the rules and guidelines that you can use when you configure multiple targets in a target connection group to write to the same Netezza target table in bulk mode:

Target Load Type	Target Options	Rules and Guidelines
Insert	Insert Update as Insert	Select the Ignore Key Constraints target property for insert targets.
Update	Update as Update Update else Insert	Use a maximum of one update table for any target. Do not use with delete tables.
Delete	Delete	Use a maximum of one delete table for any target. Do not use with update tables.

Multiple Targets Configuration for the Same Target Table

You can configure multiple targets to write to the same Netezza table, even if they are not in the same target connection group. When you configure targets in different partitions or pipelines to write to the same target table, use the same rules and guidelines as for target connection groups.

Parameterization for Netezza Targets

You can parameterize Netezza session properties to override the session properties during run time.

For example, you want to load data to a sales table. The sales table has multiple schemas to store sales information for different countries. You want to use one mapping to load data to multiple schemas based on the country name instead of creating different mappings for different schemas. In this case, you can parameterize the target table name and table name prefix in the session properties, and use the same mapping to load data to multiple schemas.

In normal mode, you can parameterize the following session properties for Netezza targets:

- Pre SQL
- Post SQL
- Reject Filename
- Table Name Prefix
- Target Table Name

In bulk mode, you can parameterize the following session properties for Netezza targets:

- Bad File Name
- Delimiter
- Error Log Directory Name
- EscapeCharacter
- NullValue
- Pipe Directory Path
- Pre SQL
- Post SQL
- Table Name Prefix
- Target Table Name

Netezza Target Data Update

In bulk mode, the PowerCenter Integration Service updates target rows based on the update options and the duplicate row handling.

Update As Insert

When you configure the session to update as insert rows, the PowerCenter Integration Service uses the following process to update target rows:

- If the source key value matches a target key value, the PowerCenter Integration Service does not insert the source row.
- If the source primary key value does not exist in the target, the PowerCenter Integration Service inserts the source row.

The following table describes how the PowerCenter Integration Service updates the target:

Source Data	Target Data	Updated Target Data	Comment
1,a,1a1	-	-	The source primary key is found in the target. The row is not inserted.
1,b,1b1	-	1,b,1b1	Inserts 1,b,1b1.
1,a,1a2	-	-	The source primary key is found in the target. The row is not inserted.
1,c,1c1	1,c,1c1	1,c,1c1	The source primary key is found in the target. The existing row 1,c,1c1 is retained. No insert is required.
1,d,1d1	-	1,d,1d1	Inserts 1,d,1d1.
1,a,1a3	1,a,1a3	1,a,1a3	The source primary key is found in the target. The existing row 1,a,1a3 is retained. No insert is required.
Note: In the pair of values, the first two values are the primary key, for example 1 (primary key), a (primary key), 1a1. The session is configured to consider key constraints.			

Update Else Insert

When you configure the session to update else insert rows, the PowerCenter Integration Service uses the following process to update target rows:

- If the source key value matches a target key value, the PowerCenter Integration Service updates *each* target row. It updates with the first or last source row matched, based on how you configure duplicate row handling.
- If the source primary key value does not exist in the target, the PowerCenter Integration Service inserts the source row.

The following table describes how the PowerCenter Integration Service updates the target:

Source Data	Target Data	Updated Target Data	Comment
1,2	1,6	1,2	Updates 1,6 with 1,2. The source primary key is found in the target. The target row is updated based on duplicate row handling to use first row.
1,3	1,8	1,2	Updates 1,8 with 1,2. Duplicate row handling is configured to update with first source row. Subsequent target rows with primary key “1” are updated with first source row.
2,4	-	2,4	Inserts 2,4. The source primary key is not found in the target. The row is inserted.
2,5	-	-	Drops 2,5. The source primary key is found in the target, and first duplicate row has been updated in the target.

Source Data	Target Data	Updated Target Data	Comment
-	3,7	3,7	Retains 3,7. No update required.
Note: In the pair of values, the first value is the primary key, for example 1 (primary key), 2.			

Netezza Session Configuration for Optimal Performance

You can increase the performance of PowerExchange for Netezza by setting the properties in the session. Set the following parameters to increase the session performance:

- Default buffer block size. You can increase or decrease the number of available memory blocks that are used to hold the source and target data in the session.
- Line sequential buffer length. You can improve the session performance by setting the number of bytes the PowerCenter Integration Service reads per line.
- Commit interval. You can increase or decrease the value of commit interval to determine the point at which the PowerCenter Integration Service commits data to the target.
- DTM buffer size. You can increase or decrease the value of DTM Buffer Size to specify the amount of memory the PowerCenter Integration Service uses as DTM buffer memory.
- Socket buffer size. You can configure the socket buffer size to specify the size of the buffers used to extract data from and load data to Netezza.
- Escape characters. You can improve session performance by avoiding the use of escape characters in a session.
- Ignore key constraints. You can improve session performance by ignoring key constraints when writing to Netezza targets. Since Netezza does not enforce key constraints, the PowerCenter Integration Service performs additional processing when a session that writes to Netezza requires key constraints.

For example, to obtain optimal performance for 3 million rows and 32 KB row size, set the following parameters:

- Default buffer block size: 1,280,000
- Line sequential buffer length: 202,400
- Commit interval: 200,000
- DTM buffer size: 28,000,000
- Socket buffer size: 8388608 bytes
- Escape characters: None
- Ignore key constraints: Selected

Netezza Distribution Key

Use a Netezza distribution key to increase session performance with parallel processing. Netezza uses a distribution key to distribute data for processing. By default, the distribution key is the first column of a table.

You can configure the distribution key to include up to four columns in a database table. When you configure a distribution key to evenly distribute data across available data slices, you can greatly increase session performance. For more information, see the Netezza documentation.

Troubleshooting Netezza Sessions

A Netezza session stops responding with no definite error messages in the logs

A Netezza session can stop responding because of the following reasons:

- The source data contains special characters like delimiter.
  A Netezza Reader session can stop responding if the source contains special characters like delimiter. Add escape characters in the session to eliminate the delimiters. This is a Netezza issue and the reference number is SWS-40577.
- The environment variables are set incorrectly.
  Check whether the environment variables PATH, LIBPATH, ODBCINI, and NZ_ODBC_INI_PATH are set correctly.
- The permissions for the file paths in the session properties are set incorrectly.
  Ensure that all the file paths in the session properties set for Netezza reader and writer sessions are correct and have proper permission. Check out for the ones which need directory path specification.

If the issue persists, you can try killing the blocking Netezza sessions or disable Netezza ODBC tracing and ODBC tracing.

How can I kill blocking Netezza sessions?

Use the nzsession utility, which comes with client tools, to kill blocking Netezza sessions.

Run the following command to view the active Netezza sessions:

```
nzsession show -host <hostname> -u <user> -pw <password> -maxColW <column width> |grep -
i "active
```

Run the following command to kill active sessions:

```
-host <hostname> -u <user> -pw <password> -id <session id> [-force]
```

How can I enable or disable Netezza ODBC tracing?

In the odbcinst.ini file, set the parameter debugLogging as true to enable Netezza ODBC tracing and as false to disable Netezza ODBC tracing.

How can I enable or disable ODBC tracing?

In the odbc.ini file, set the parameter Trace as 1 to enable Netezza ODBC tracing and as 0 to disable Netezza ODBC tracing.

APPENDIX A

Datatype Reference

This appendix includes the following topic:

- Netezza and Transformation Datatypes, 28

Netezza and Transformation Datatypes

PowerCenter uses the following datatypes in Netezza mappings:

- Netezza native datatypes. Netezza datatypes appear in Netezza definitions in a mapping.
- Transformation datatypes. Set of datatypes that appear in the transformations. They are internal datatypes based on ANSI SQL-92 generic datatypes, which the PowerCenter Integration Service uses to move data across platforms. They appear in all transformations in a mapping.

When the PowerCenter Integration Service reads source data, it converts the native datatypes to the comparable transformation datatypes before transforming the data. When the PowerCenter Integration Service writes to a target, it converts the transformation datatypes to the comparable native datatypes.

The following table lists the Netezza datatypes that PowerCenter supports and the corresponding transformation datatypes:

Netezza Datatype	Range	Transformation Datatype	Range
BigInt	Precision 19, scale 0	Bigint	From -9,223,372,036,854,775,808 through 9,223,372,036,854,775,807 Precision of 19, scale of 0 Integer value
Bool	True or false, on or off, 0 or 1, yes or no	String	Precision 1
ByteInt	Precision 3, scale 0	Small Integer	Precision 5, scale 0
Char	Single character	String	From 1 through 104,857,600 characters
Date	ANSI SQL date	Date/Time	Jan. 1, 0001 A.D. to Dec. 31, 9999 A.D. (precision to the nanosecond)
Float8	Precision 15	Double	Precision 15
Float4	Precision 6, scale 0	Double	Precision 15

Netezza Datatype	Range	Transformation Datatype	Range
Integer	Precision 10, scale 0	Integer	Precision 10, scale 0
NChar(m)	Single character Used for storing UTF-8 data	nString	From 1 through 104,857,600 characters
NVarchar(m)	BVarchar (length) Non-blank-padded string, variable storage length Used for storing UTF-8 data	nString	From 1 through 104,857,600 characters
Numeric	Numeric (precision, decimal), arbitrary precision number Precision must be between 1 and 38	Decimal	Precision from 1 through 28 digits, scale from 0 through 28
Real	Precision 6, scale 0	Real	Precision of 7, scale of 0 Double-precision floating-point numeric value
SmallInt	Precision 5, scale 0	Small Integer	Precision 5, scale 0
Time	hh:mm:ss. ANSI SQL time	Date/Time	Jan. 1, 0001 A.D. to Dec. 31, 9999 A.D. (precision to the microsecond)
Timestamp	Precision 26, scale 6	Date/Time	Jan. 1, 0001 A.D. to Dec. 31, 9999 A.D. (precision to the microsecond)
Varchar	Varchar (length) Non-blank-padded string, variable storage length	String	From 1 through 104,857,600 characters

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